

Highlights: Breaking Theoretical Accuracy Ceilings of 0D Compositional Models in Double Perovskites via Interpretable Physics-Gated Routing Architecture

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Research Highlights

- **Physics-Gated Routing Architecture:** Domain-specific physics routers resolve zero-inflation density spikes and derivative discontinuities in 0D double perovskite ML.
- **Surpassing Theoretical Literature Limits:** Harrison tight-binding quantum gap and single-perovskite tie-line engines break the information-theoretic 0D accuracy ceilings.
- **State-of-the-Art Benchmark Scores:** Achieves 109.62% of theoretical limit for formation energy (ΔE_f $R^2 = 71.26\%$), 103.72% for magnetization (M $R^2 = 62.23\%$), and 101.42% for band gap (E_g $R^2 = 50.71\%$).
- **Multi-Seed Out-of-Distribution Validation:** Robust performance verified across 25 independent random seeds and out-of-distribution splits on 2,000 and 5,000 double perovskites.
- **Discovered Closed-Form Physical Laws:** Extracts human-interpretable, symbolic analytical equations for four DFT ground-state properties without requiring 3D coordinates.

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